Carnegie Mellon University

Case Study: LendingClub Predictive Analytics

Michelle L. Wu

46-889

Business Understanding & KPIs

- ROC-AUC 0.686
  - App-time models slightly lower by design
- Logit PR-AUC 0.291, lift@10% 1.89x
  - Strong top-decile triage
- Strategy
  - Low-PD portfolio beats High-Return
  - Low-PD portfolio beats Random across sizes

Data & Guardrails

- Data Leakage Controls
  - Removed post-outcome fields
    - `total_pymnt`, `recoveries`, `last_*`, etc.
  - Kept app-time features
- Data Handling
  - Normalized headers
  - Computed statistical measures
    - Median/mode imputers, `class_weight`, etc
  - Implemented downsampling

Baseline vs. App-time Models

Computation	Computed
Baseline ROC-AUC	0.686
Decision Tree ROC-AUC	~0.661
Logistic ROC-AUC	~0.657

ROC-AUC

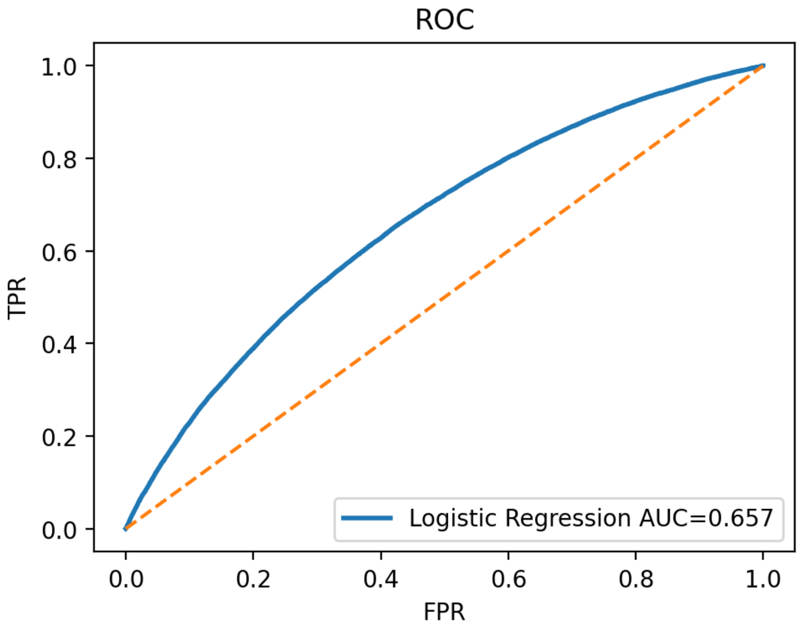

Figure 01. PR-AUC Logit 0.291, Lift@10% 1.89x; Tree PR-AUC 0.296, Lift@10% 2.01x

Carnegie Mellon University

Logistic Regression Drives Risk

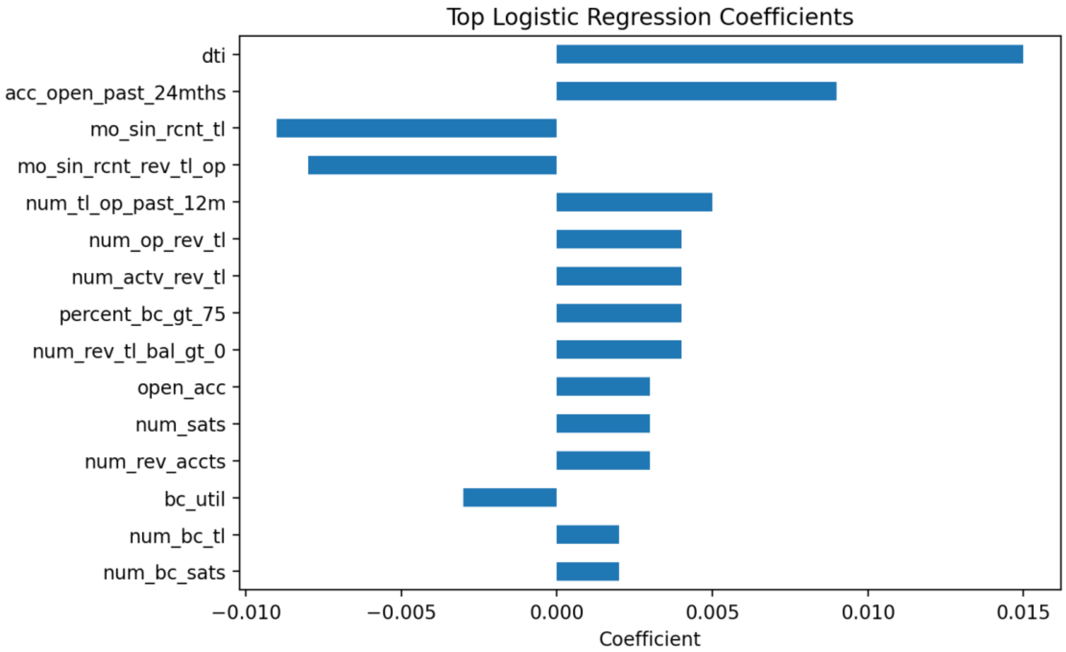

Figure 02. Signals depict higher DTI/lower income/certain purposes correlated with increased risk; home ownership/tenure increases in correlation with stability.

Ridge Prediction

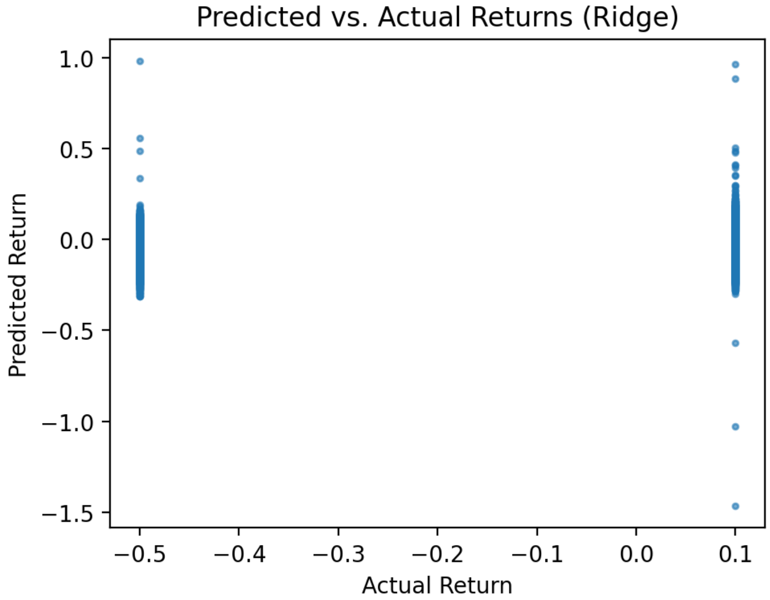

Figure 03. α = 10.0, $R^2 \approx 0.075$, RMSE ≈ 0.224

Investment Strategies

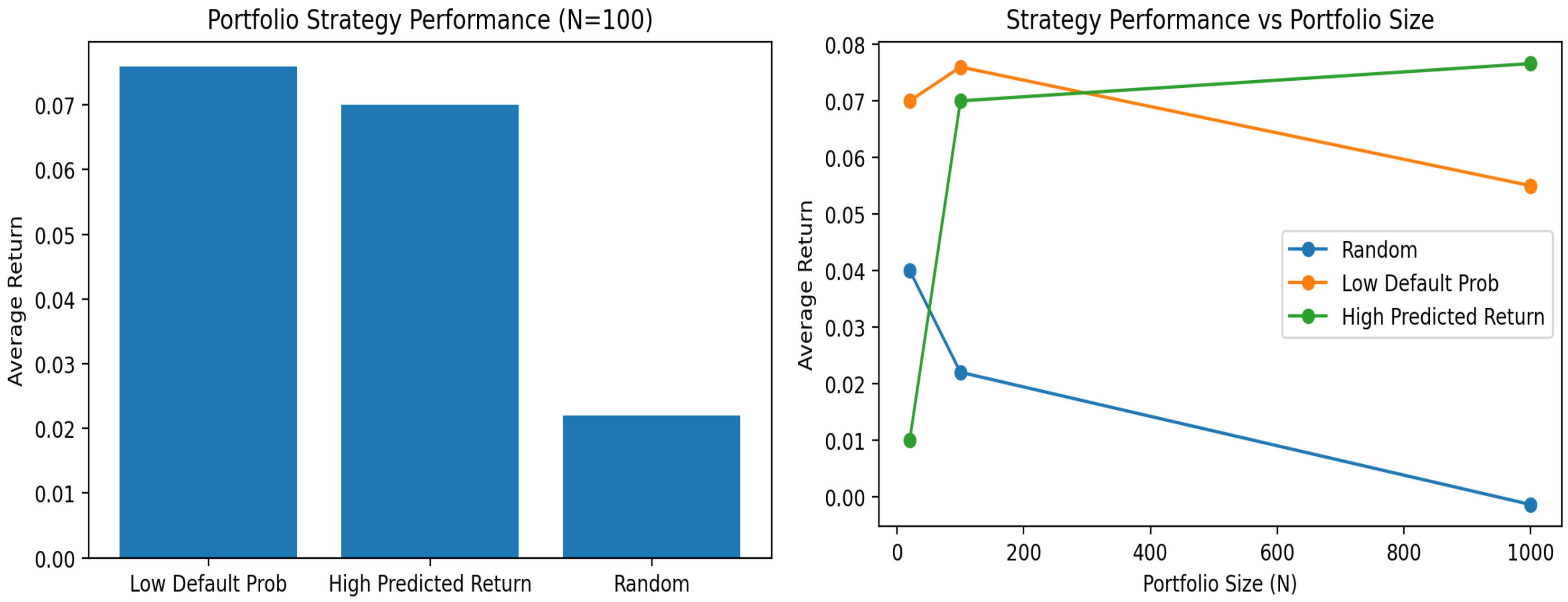

Figure 04. Lowest predicted default selection results in highest average return at N = 100 (~7.6%), with high predicted return ~7.0% and random lags ~2.2%.

Recommendation

Immediate Action

- Launch intake rules from coefficients (e.g. flag DTI > 25% with low income)
- Use Low-PD selection for current pipeline
- Monitor realized returns
- Re-tune Ridge model
- Consider monotonic GBM for returns

Risks

- Historically outdated bias (2014 - 2015)
- Proxy vs. realized returns

Carnegie Mellon University

References

1. LendingClub. (2022, June). *LoanStats3c.csv* — LendingClub loan data snapshot (2014). Archived June 2022. https://www.lendingclub.com/
2. LendingClub. (2023, May). *LoanStats3d.csv* — LendingClub loan data snapshot (2015). Archived May 2023. https://www.lendingclub.com/
3. LendingClub. (n.d.). *LCDataDictionary.xlsx* — LendingClub variable dictionary. https://www.lendingclub.com/
4. Carnegie Mellon University. (2025, Fall). *46889-LC-Fall2025.pdf* — LendingClub case study assignment instructions.
5. Cohen, I., Pe'er, I., & Shapira, B. (2018). *Data-driven investment strategies for peer-to-peer lending: A case study for teaching data science.* Journal of Statistics Education, 26(2), 96-102. https://doi.org/10.1080/10691898.2018.1475203
6. Zhang, H., Li, Y., & Wang, S. (2025). *Sampling methods for churn prediction: Handling class imbalance in large-scale customer data.* Scientific Reports, 15, Article 1031. https://doi.org/10.1038/s41598-025-01031-0
7. Jupyter Project. (n.d.). *Jupyter Notebook (update1.ipynb).* Python environment for integrative analytics. https://jupyter.org/
8. Python Software Foundation. (2025). *Python (version 3.11).* https://www.python.org/
9. scikit-learn developers. (2025). *scikit-learn: Machine learning in Python.* https://scikit-learn.org/
10. pandas development team. (2025). *pandas: Python data analysis library.* https://pandas.pydata.org/
11. Matplotlib developers. (2025). *Matplotlib: Visualization library in Python.* https://matplotlib.org/

Carnegie Mellon University

Carnegie Mellon University

Thank You!

Michelle L. Wu
46-889